

Credit Card Fraud Detection

End-to-End Data Analytics — Python · SQL · Power BI

BANKING / FINANCIAL SERVICES

284,807 TRANSACTIONS



Made with GAMMA

Executive Summary

284,807

Total Transactions

492

Fraud Transactions

0.17%

Fraud Rate

\$25.16M

Total Amount

Severe class imbalance — fraud is rare but valuable to detect.

Dashboard: Executive + Fraud Analytics pages.



Business Problem & Objectives



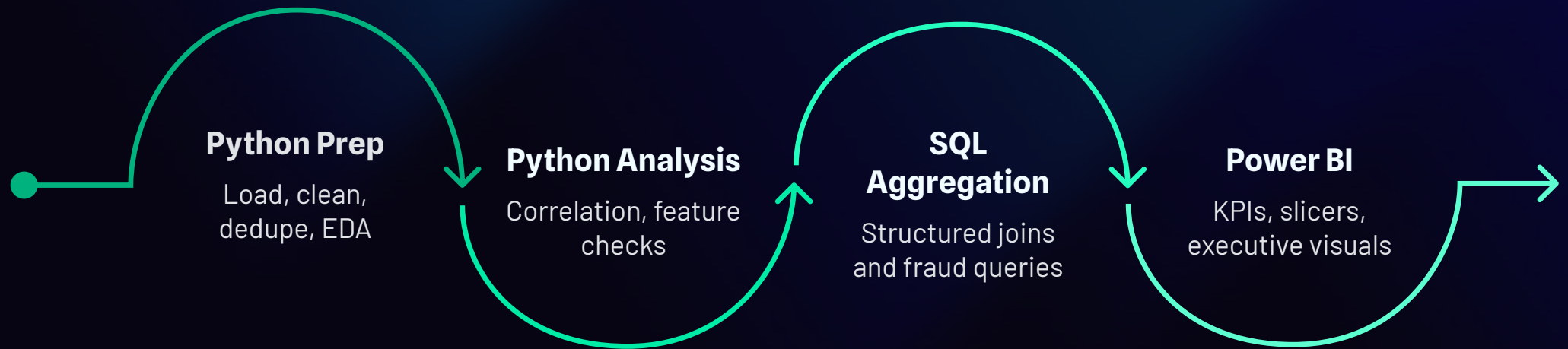
Core Questions

- Fraud rate & time variation
- Transaction value behavior
- PCA feature associations

Objectives

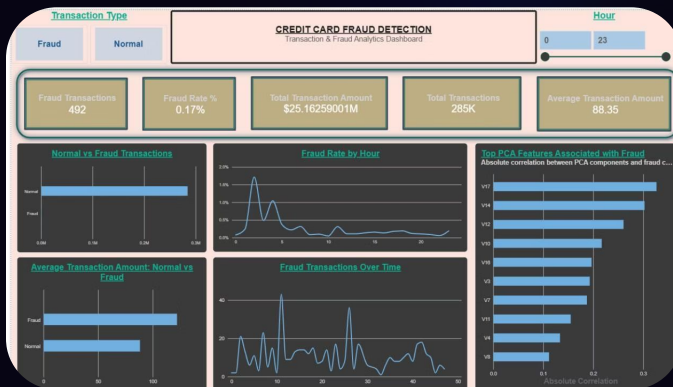
- Clean & validate dataset
- Python EDA + SQL analysis
- Power BI dashboards
- Fraud-monitoring recommendations

Dataset & Methodology



31 columns: Time, V1-V28 (PCA-transformed), Amount, Class. 1,081 duplicates removed. **Note:** Time = elapsed seconds, not calendar timestamp.

Executive Dashboard

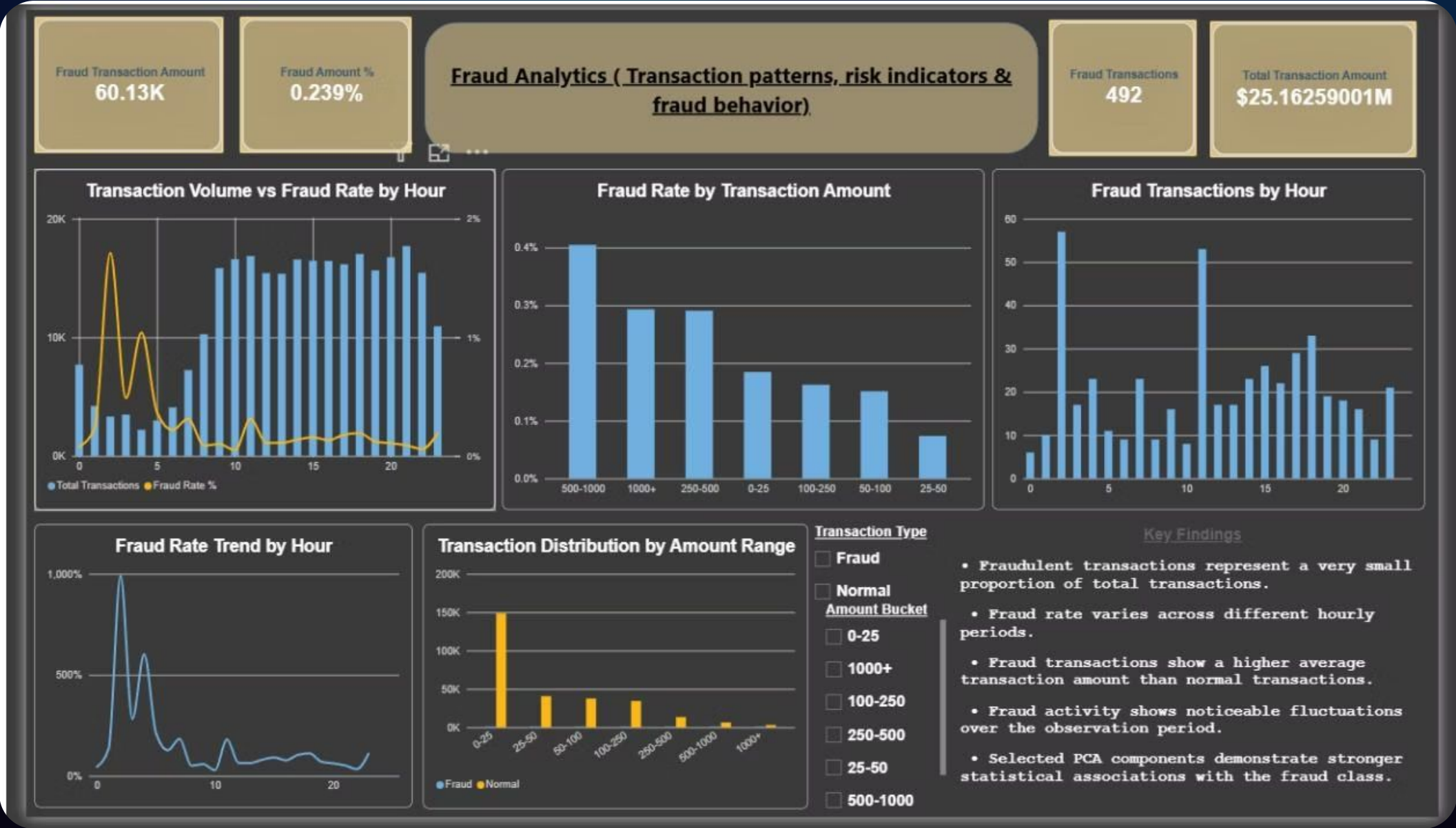


Key Visuals

- Normal vs. Fraud transaction bar chart
- Fraud rate trend by hour
- Top PCA features by correlation
- Average amount: Normal vs. Fraud

❏ Fraud transactions show **higher average amount** than normal transactions.

Fraud Analytics Dashboard



\$60.13K

Fraud Amount

0.239%

Fraud Amount %

Peak Hours

Hour 2 & Hour 12

Highest Risk

\$500-\$1,000 bucket

Detailed Fraud Findings

Financial Exposure

Fraud = **0.17%** of volume but **0.239%** of value — fraud is disproportionately high-value.

Amount Ranges

\$500–\$1,000 bucket shows highest fraud rate, followed by \$1,000+ and \$250–\$500.



Business Recommendations

1

Multi-Signal Scoring

Combine amount, time, PCA features into a risk score.

2

Monitor Fraud Spikes

Automate alerts for sudden fraud rate increases.

3

Right Metrics

Precision, Recall, F1, ROC-AUC – not accuracy.

4

Power BI Monitoring

Management dashboard + Python/ML risk scores in production.



Technical Implementation

Python

- Pandas: load, clean, EDA
- Correlation: rank PCA features

Power BI

- DAX: Fraud Rate %, Fraud Amount %
- Calculated columns: Hour, Amount Bucket
- Interactive slicers

Measure	Purpose
Total Transactions	Count all rows
Fraud Rate %	Fraud ÷ Total
Fraud Amount %	Fraud Amt ÷ Total Amt

Limitations & Next Steps

Current Limitations

- V1-V28: anonymized PCA — no business labels
- Time = elapsed seconds, not calendar
- No merchant, device, or location context

Next Phase

- Logistic Regression & Random Forest models
- Stratified split + imbalance techniques
- Page 3: ML Performance (Confusion Matrix, ROC-AUC)